Julian Gross

Internship | Quantum Algorithms & Simulation

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Physics MSc student at ETH Zürich with a strong focus on scientific computing, numerical simulation, and classical methods for analyzing quantum algorithms. Experienced with Majorana Propagation, Pauli Propagation, tensor network, matrix product state simulations, and benchmarking of quantum circuits. Seeking a research-oriented internship to contribute to and learn about quantum algorithm development, simulation, and performance analysis.

Education

ETH Zürich, MSc Physics Sept 2022 – today

Relevant coursework

- **Computational Quantum Computing :** MPS, exact diagonalization, Variational and Determinant Quantum MC, Path Integrals, Neural Network Quantum states, finite-difference ODE/PDE solvers
- **Quantum Information and systems** : Bosonic QIT (binomial, 2-component cat, GKP codes), Open Quantum Systems, Input-Output Theory, Quantum Acoustics and Optomechanics
- **Numerical Methods :** Fourier methods, nonlinear least-square fitting, eigenvalue algorithms, Feed-Forward Neural Networks, Stochastic Gradient Descent, Stochastic Reconfiguration

Projects

Research Project: Majorana Propagation, Institute for Theoretical Physics ETH Zürich Feb – Aug 2026

- Simulated Fermi-Hubbard dynamics on a 7 × 4 torus using MajoranaPropagation.jl with dual truncation schemes to replicate trapped-ion quantum computer data of arxiv.org/abs/2510.26300
- Applied exact diagonalization in netket simulations to verify methods on smaller system sizes
- Benchmarked trapped-ion hardware results, identifying precision limits and exact boundary regimes where the method breaks down
- Optimized truncation algorithm to increase calculation accuracy from 3e-5 to 1e-6

Variational Neural Network Quantum State, ETH Zürich Dez 2025 – Jan 2026

- Benchmarked Jastrow and Restricted Boltzmann Machine variational state ansatzes against exact ground state energies of the 1D modified Haldane-Shastry spin model
- Evaluated optimization convergence and energy precision to reproduce spin correlation results of Deng et al. arxiv.org/abs/1701.04844

Proseminar: Classical Simulations of Quantum Computing, ETH Zürich Sept – Nov 2025

- Conducted an independent research project on classical simulation of quantum circuits using tensor network methods
- Studied Belief-Propagation Tensor Network States in the Vidal gauge, applied to the classical analysis of IBM's Eagle kicked Ising experiment
- Presented a one-hour technical seminar to peers and faculty, explaining tensor network concepts and benchmarking methodology

Technical skills

- Programming: Python, Julia, C++
- Scientific computing: SciPy, Pandas, Scikit-learn, ITensors, NetKet, MajoranaPropagation
- Algorithms and methods: Monte Carlo Methods, Belief-Propagation, MPS, Pauli-based propagation
- Data analysis and simulation : Time-series and spectral analysis, regression, basic machine learning

Professional Experience

Mathematics teacher, freieMaturitätsschule Winterthur, ZH Sept 2024 – Feb 2025

- Educated a class of high school students in linear algebra, stochastics and analysis, delivered lessons while balancing a full-time course load at ETH, developed multiple exams to grade students

Achievements

- 10-year career as a Rock'n'Roll athlete, 4 times national champion, 11th place at the 2021 European Championships in Kharkiv, 5-year career as an Aerobic Gymnastic athlete, 4 times national champion